Republic of the Philippines
CAVITE STATE UNIVERSITY
Don Severino de las Alas Campus
Indang, Cavite

SELF-ASSESSMENT CHECKLIST
Research Proposal Making for Computer Science

Instruction: Review the self-assessment criteria provided and evaluate your CS research proposal. Please denote whether each criterion has been satisfied with either a '*Yes*' or '*No*' within the self-assessment form by putting a checkmark (✓) in the appropriate box accordingly.

Proposed Thesis Title : **A Cross-Layer Network Diagnostic System for Identifying Performance Bottlenecks in Local Area Networks using Heuristic Analysis**
Area of Study : **Artificial Intelligence**

NO.	CRITERIA	YES	NO
1	Is your research proposal clearly defined and aligned with the research tracks and program objectives of computer science?	✓	
2	Does your research aim to address a gap in existing knowledge?	✓	
3	Does your research propose a novel/original solution to a CS-related problem?	✓	
4	Does your proposed methodology involve the creation and analysis of computer models to simulate complex systems or phenomena?	✓	
5	Are you going to (develop new **OR** modify existing) algorithms or software to imitate processes, events, or behaviors?	✓	
6	Does your methodology include hands-on experimentation, data collection, and analysis to validate hypotheses or test new ideas?	✓	
7	Are you implementing and testing your ideas in real-world scenarios or through practical applications? **NOTE:** This pertains to assessing the quality or attributes of the software through testing rather than gathering opinions from individuals regarding their perception of the software.	✓	
8	Does your thesis primarily focus on developing new concepts, frameworks, or models without extensive experimentation or implementation?	✓	
9	Does your proposed methodology rely on mathematical proofs, logic, or conceptual discussions to support your research findings?	✓	
10	Will your research contribute new insights, theories, algorithms, or methodologies to the broader field of computer science?	✓	
11	Will your proposed research be considered acceptable and relevant by the academic community and potential stakeholders?	✓	
12	Does your proposal have direct client or organization who will serve as your beneficiary?	✓	
13	Does your proposal will involve database and web/mobile app?	✓	
14	Does your research proposal align with accepted research areas within the field of computer science, such as artificial intelligence, data science, algorithms, computer networks, etc.?	✓	
12	Overall, is this proposal acceptable if you were the evaluator?	✓	

Submitted by:

Balba, Terrence Joshua L.
Name and Signature of the Student

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Date

RESEARCH PROPOSAL TITLE

Proponents

Main Proponent: Balba, Terrence Joshua L.

Group Members: Soliman, Daniella Lyn G.

Padillo, Lance Jay

Involved Sustainable Development Goals

1. SDG 9 – Industry, Innovation and Infrastructure

Involved CvSU Research Thematic Area

1. Orange Area - Smart Engineering, ICT and industrial competitiveness

Rationale

The dependence on a stable and fast internet connection for remote working, education, and digital collaboration has made the Local Area Network (LAN) an essential tool in our lives. Nevertheless, as the complexity of home and office networks grows, it is a major challenge for the average end-user to pinpoint the actual cause of network performance degradation. In cases of poor network performance, an average end-user typically employs a number of commonly available internet speed test tools to diagnose the problem. These tools, however, have a major drawback in that, while they can measure end-to-end network performance, they cannot trace the performance bottlenecks to their constituent parts of the network path. This means that an end-user cannot pinpoint whether a poor network performance is due to an Internet Service Provider (ISP) constraint, wireless interference from the router, or a hardware constraint on their personal device. Recent studies have directly observed this phenomenon, stating that for users who have a high-speed internet connection, the actual performance bottleneck lies in their local wireless network 100% of the time (Sharma et al., 2023).

Identifying such network faults in a particular network is a complex task in itself. Today's network infrastructures face localized congestion, hardware constraints, and other issues that impact network performance, but identifying such network faults is a notoriously difficult task because of the absence of appropriate intranetwork observability (Kong et al., 2023). Today's network fault identification techniques, which use deep learning and sophisticated neural networks, have been proposed to identify network anomalies, but such techniques require enormous memory, computational power, and cloud connectivity. For localized fault identification, using machine learning techniques is practically infeasible, but it has been proven that using decentralized techniques to identify network faults using heuristics and optimization rules is highly successful, requiring much less computational power and memory compared to machine learning techniques (Tuli et al., 2022).

To address this need, there is a distinct need for an accessible and intelligent diagnostic tool that is capable of automatically identifying the causes of local network problems without requiring complex computational models. By using a cross-layer heuristic approach and correlating physical-layer metrics such as WiFi signal strength with network-layer metrics such as ping delays and local hardware utilization, it is possible to accurately identify the cause of a network bottleneck. As such, the need for the development of a cross-layer network diagnostic tool using a heuristic approach for evaluating network telemetry is established. By simulating the logical deduction process undertaken by a human network technician, it is possible for a non-technical user to accurately determine the cause of internet performance problems.

Significance of the Study

The development of a cross-layer network diagnostic system utilizing heuristic analysis will provide highly practical and theoretical value. Specifically, the results of this study will be highly beneficial to the following groups:

Non-Technical End-Users (Remote Workers and Students). The main target group for the application of the findings of this study will be the average internet user who requires a consistent internet service for productive activities on a day-to-day basis but does not possess the required knowledge in IT network configuration. By offering a tool that is capable of identifying the exact cause of a network bottleneck, distinguishing between a throttle caused by the ISP, a weak router signal, or a CPU overload, the average internet user is empowered with the knowledge required to resolve the true cause of the internet problems they are experiencing.

Internet Service Providers (ISPs) and Customer Support Teams. Internet Service Providers often receive service complaints about internet speeds being slow, which is actually caused by local wireless interference, distance from the router, or the user's old hardware. By providing the end-user with a tool that is capable of distinguishing between local network faults and external access link faults, the overall number of false-positive service complaint tickets can be minimized. This enables the support teams at the Internet Service Providers to minimize costs.

IT Administrators and Helpdesk Personnel. For IT professionals dealing with a remote/hybrid workforce, it can be very difficult to figure out what an employee’s home network environment looks like. This heuristic-based system helps IT departments quickly and easily determine if a remote employee’s connectivity issues are with the hardware they have been provided with, their Wi-Fi at home, or their local internet service provider.

The Academic Community and Future Researchers. In the Computer Science domain, in the Computer Networks and Data Science tracks, this research provides a real-world application of heuristic analysis in the realm of network telemetry. It proves that a viable, cross-layer fault isolation mechanism can be achieved without the significant computational expense and cloud connectivity that more advanced machine learning techniques require. The heuristic logic developed here can be used as a base reference for future researchers seeking to optimize a lightweight network diagnostic tool or incorporate rule-based logic into a larger smart home solution space.

Objectives of the Study

General Objective The main objective of this study is to develop a cross-layer network diagnostic system that utilizes heuristic analysis to isolate and identify the specific source of performance bottlenecks within a Local Area Network (LAN).

Specific Objectives Specifically, the study aims to achieve the following:

1. **To develop cross-layer telemetry extraction modules** capable of securely retrieving local physical-layer metrics (e.g., WiFi Received Signal Strength Indicator), network-layer metrics (e.g., local gateway ping and external access link latency), and local hardware utilization metrics (e.g., CPU and network interface throughput).

2. **To design and implement a heuristic, rule-based algorithm** (an expert system) that analyzes and correlates the extracted telemetry to accurately classify the root cause of network degradation into one of three categories: Internet Service Provider (ISP) limitations, local wireless/router interference, or local hardware constraints.

3. **To integrate the diagnostic logic into a desktop application** featuring a graphical user interface (GUI) that allows non-technical users to easily execute the diagnostic tests and view actionable troubleshooting results.
4. **To evaluate the accuracy and reliability of the diagnostic system** through hands-on, controlled network fault simulations (e.g., inducing high CPU load, simulating wireless signal attenuation, and restricting access link bandwidth) to validate the system's heuristic logic in real-world scenarios.

Literature Review

Automated network performance diagnosis has seen increasing attention as networks grow in complexity and scale. Traditional performance tests (e.g., throughput or latency measurements) help quantify end-to-end behavior but do not explain *why* the performance is degraded at specific layers of the network stack.

Modern research trends can be grouped into three major approaches:

1. Anomaly Detection with Expert Knowledge Integration

Recent work highlights that combining statistical anomaly detection with domain knowledge improves human-aligned troubleshooting. Navarro *et al.* (2021) proposed a hybrid system that integrates unsupervised anomaly detection with an expert knowledge module to interpret anomalies from router telemetry data, showing high agreement with human diagnosis and improved troubleshooting performance.

2. Machine Learning for Network Fault Prediction

Machine learning models have been widely applied to network reliability and maintenance tasks. For instance, Rzayeva *et al.* (2023) developed machine learning–based prediction models (Decision Trees, SVM, etc.) for LAN equipment failure, demonstrating that pattern-based predictors can anticipate faults with high accuracy. Separately, a 2024 study combined double exponential smoothing with ML algorithms to improve the prediction of LAN switch failures, indicating the value of hybrid forecasting and classification techniques in preemptive network health management.

Despite their effectiveness in prediction, these approaches target *forecasting failures* rather than *diagnosing root causes* across network layers, and often require extensive training data.

3. Knowledge and Data Fusion Diagnostics in Complex Networks

In larger and more heterogeneous systems, fault diagnosis has started to use knowledge and data fusion techniques. Zhao et al. (2024) presented a fault diagnosis method for 5G cellular networks that incorporates expert knowledge and uses GANs and graph neural networks to understand multi-source indicators and fault type diagnosis with high accuracy.

Tan et al. (2025) have also proposed a framework using large language models and symbolic representations for multi-scenario network diagnosis, demonstrating the effectiveness of semantic representations and updating knowledge for fault resolution generalization.

Approach	Strength	Limitation
Anomaly + Knowledge Integration	Aligned with human reasoning	Limited root-cause detail
Machine Learning Prediction	High accuracy given data	Needs large datasets; less interpretable
Knowledge-Driven & Hybrid Models	Interpretability + multi-source fusion	Often complex and domain specific

Gap in Current Literature

Most existing studies either:

- Detect anomalies without interpreting cross-layer causes; or
- Forecast failures rather than *explain* their origins; or
- Apply complex models that are resource-intensive and hard to deploy on edge devices.

There is **limited research on lightweight, rule-based diagnostic frameworks** that apply domain knowledge to correlate cross-layer network indicators and produce interpretable root-cause classifications for LAN performance degradation.

This gap motivates the proposed research: a **heuristic, expert-style diagnostic model** that synthesizes multiple performance metrics and applies rule-based reasoning to classify network bottlenecks, providing explainable insights with minimal data and computational requirements.

Methodology

Research Design

This study adopts a **design and experimental research approach**. The research involves:

1. Designing and implementing a cross-layer heuristic-based diagnostic system.
2. Conducting controlled fault simulations.
3. Evaluating classification performance using standard diagnostic metrics.

The methodology focuses on empirical validation through controlled experimentation rather than survey-based evaluation.

System Architecture

The proposed system consists of four major components:

1. Telemetry Extraction Module
2. Feature Normalization and Preprocessing
3. Heuristic Inference Engine (Expert System)
4. Graphical User Interface (GUI)

Telemetry Feature Extraction

The diagnostic system that is being proposed uses quantitative telemetry data that is gathered from different layers of the network stack. These quantitative telemetry data will then be used by the heuristic inference engine to identify what is probably causing the network performance degradation. The telemetry data that is being extracted includes physical layer signal metrics, network layer latency indicators, as well as hardware utilization metrics from the client device.

In order for the extracted telemetry data to be consistent and comparable, different statistical calculations will be performed on the extracted telemetry data.

A. Average Network Latency

a. Network latency is measured through multiple Internet Control Message Protocol (ICMP) ping tests to both the local gateway and an external server (e.g., a public DNS server). The system computes the average latency using the following formula:

$$L_{avg} = \left(\frac{1}{n}\right) * \sum L\_i$$

where:

- L_i represents the latency measured during the i^{th}ping attempt
- n represents the total number of ping samples collected

The difference between external latency and gateway latency provides an indicator of whether the delay originates from the local network or the external Internet connection.

B. Wireless signal strength normalization

a. The wireless signal strength between the client device and the router is measured using the Received Signal Strength Indicator (RSSI), expressed in decibel-milliwatts (dBm). To standardize the signal measurements, RSSI values are normalized using the following equation:

$$RSSInorm = \frac{RSSI - RSSImin}{RSSImax - RSSImin}$$

where:

- $RSSI$ represents the measured signal strength
- $RSSI_{min}$ represents the minimum expected signal level
- $RSSI_{max}$ represents the maximum expected signal level

Signal strength values below −70 dBm are considered weak and may indicate potential wireless interference or excessive distance between the client device and the access point.

C. CPU Utilization Measurement

a. Hardware constraints are evaluated by monitoring the CPU utilization of the client device during diagnostic testing. CPU usage is computed using the following equation:

$$CPU_{usage} = \frac{CPU_{busy}}{CPU_{total}} * 100$$

where:

- CPU_{busy} represents the time during which the processor is actively executing tasks
- CPU_{total} represents the total observation time

CPU utilization exceeding a predefined threshold (e.g., 80%) may indicate that the device hardware is contributing to reduced network performance.

Heuristic Diagnostic Model

The diagnostic part of the system uses a heuristic rule-based inference model to simulate the thought process of a human network technician. Unlike the computationally intensive approach of machine learning algorithms, the system uses expert rules to correlate the telemetry metrics collected from the various network layers.

The diagnostic logic classifies the network bottleneck into one of three categories:

1. Internet Service Provider (ISP) limitation
2. Wireless or router interference
3. Local hardware constraint

Each diagnosis is determined by evaluating network indicators against predefined heuristic rules.

Wireless Interference Rule

Wireless interference is suspected when the signal strength is weak and local latency is elevated.

Example rule:

IF RSSI < −70 dBm
AND gateway latency exceeds threshold
THEN network bottleneck = Wireless Interference

Hardware Constraint Rule

Hardware limitations are suspected when device resource utilization is high while network conditions remain normal.

Example rule:

IF CPU usage > 80%
AND network throughput is low
THEN network bottleneck = Hardware Constraint

ISP Bottleneck Rule

An ISP limitation is suspected when the local network conditions are healthy but external network latency is high.

Example rule:

IF RSSI ≥ −60 dBm
AND CPU usage < 70%
AND external latency is significantly higher than gateway latency
THEN network bottleneck = ISP Limitation

These rules allow the system to correlate physical-layer metrics, network-layer measurements, and device performance indicators in order to determine the most probable cause of network degradation.

Experimental Setup

To validate the system, controlled fault scenarios will be created.

1. Test Environment
2. Controlled Fault Scenarios

   Each bottleneck category will be simulated independently.

   a. **Scenario A: ISP Bottleneck Simulation**
      i. Artificial bandwidth restriction via router QoS
      ii. External latency artificially increased

      Ground Truth Label: ISP Bottleneck

   b. **Scenario B: Wireless Interference Simulation**
      i. Increase distance from router
      ii. Introduce physical obstructions
      iii. Change WiFi channel to congested band

      Ground Truth Label: Wireless Interference

   c. **Scenario C: Hardware Constraint Simulation**
      i. Run CPU stress program
      ii. Introduce background heavy applications

      Ground Truth Label: Hardware Constraint

Each scenario will be repeated multiple times (e.g., 30 trials per category) to generate sufficient evaluation data.

Evaluation Metrics

For the evaluation of the effectiveness of the proposed diagnostic system, standard performance metrics are used. These metrics will assess the accuracy of the heuristic diagnostic model in determining the source of network bottlenecks.

Accuracy

Accuracy measures the proportion of correctly classified diagnostic outcomes relative to the total number of experimental trials.

Accuracy = (TP + TN) / (TP + TN + FP + FN)

where:

- TP represents true positive classifications
- TN represents true negative classifications
- FP represents false positive classifications
- FNrepresents false negative classifications

Precision

Precision evaluates how many predicted bottleneck classifications are actually correct.

$$Precision = \frac{TP}{TP + FP}$$

Recall

Recall measures the ability of the system to correctly detect actual bottleneck conditions.

$$Recall = \frac{TP}{TP + FN}$$

F1 Score

The F1-score provides a balanced metric that combines both precision and recall.

$$F1 = \frac{2(Precision \times Recall)}{Precision + Recall}$$

Statistical Validation

To ensure reliability:

- Each scenario will be tested with at least 30 independent trials.
- Mean accuracy and standard deviation will be computed.
- Results will be compared against baseline random classification (33% accuracy for 3 classes).

The variance of the classification accuracy across trials is computed as follows:

$$s^2 = \left(\frac{1}{n-1}\right)\sum\left(x_i - \bar{x}\right)^2$$

where:

- $x_i$represents the accuracy value observed in trial i
- $\bar{x}$represents the mean accuracy across all trials
- $n$represents the total number of trials

Lower variance values indicate that the diagnostic system produces consistent and stable results across repeated experiments.

Ethical Considerations

- No personal user data will be collected.
- Only system-level telemetry is used.
- No traffic payload inspection is performed.

Expected Output

Upon completion of the study, the following outputs are expected:

1. Functional Cross-Layer Network Diagnostic System

The primary output of this research will be a fully functional desktop-based diagnostic application that:

- Collects real-time telemetry from multiple network layers

- Applies a heuristic rule-based inference engine
- Classifies performance bottlenecks into:
    - ISP Limitation
    - Wireless/Router Interference
    - Local Hardware Constraint
- Provides user-readable diagnostic explanations
- Recommends appropriate troubleshooting actions

The system will operate without reliance on cloud services or heavy machine learning models, ensuring lightweight deployment suitable for consumer and small-office environments.

2. Implemented Heuristic Knowledge Base

The research will produce a structured and documented:

- Set of production rules (IF–THEN logic)
- Threshold definitions for diagnostic indicators
- Cross-layer reasoning framework

This knowledge base will serve as a reusable reference for future researchers developing lightweight diagnostic systems.

3. Experimental Performance Results

The study will produce empirical results based on controlled fault simulations, including:

- Classification Accuracy
- Precision per bottleneck category
- Recall per bottleneck category
- F1-Score
- Confusion Matrix

Expected outcome:

The heuristic model is expected to achieve classification accuracy significantly higher than baseline random classification (33% for three classes). A target performance threshold of **≥80% overall accuracy** is anticipated under controlled testing conditions.

4. Diagnostic Time Performance Metrics

The system is expected to:

- Complete full diagnostic evaluation within 5–10 seconds
- Perform inference in near real-time (low computational overhead)
- Maintain CPU usage below a defined lightweight threshold (e.g., <10% during diagnostic execution)

This validates the efficiency advantage over heavier machine learning models.

5. Comparative Insight Against Existing Tools

The study will demonstrate that:

- Traditional speed tests provide throughput metrics but not root-cause classification.
- The proposed system provides structured diagnostic interpretation.
- The heuristic approach offers explainability that purely statistical models may lack.

6. Academic Contributions

The expected theoretical contributions include:

- A cross-layer heuristic diagnostic framework
- A structured methodology for evaluating rule-based network diagnosis
- Empirical evidence supporting lightweight expert-system approaches in LAN environments

REFERENCES

Kong, X., Lou, J., Bai, W., Nam Sung Kim, & Zhuo, D. (2023). *Towards a Manageable Intra-Host Network*. https://doi.org/10.1145/3593856.3595890

Li, Y., Qu, J., Xin, K., & Wang, Z. (2024). Research on Application Methods of Network Fault Prediction Models. *World Scientific Research Journal*, *10*(8), 83–89. https://doi.org/10.6911/WSRJ.202408_10(8).0010

Navarro, J. M., Huet, A., & Rossi, D. (2021). *Human readable network troubleshooting based on anomaly detection and feature scoring*. ArXiv.org. https://arxiv.org/abs/2108.11807

Rzayeva, L., Myrzatay, A., Gulnara Abitova, Assiya Sarinova, Korlan Kulniyazova, Saoud, B., & Ibraheem Shayea. (2023a, October 13). *Anomaly Detection Algorithms for LAN Failure Prediction*. Encyclopedia.pub; Leila Rzayeva. https://encyclopedia.pub/entry/50242

Rzayeva, L., Myrzatay, A., Gulnara Abitova, Assiya Sarinova, Korlan Kulniyazova, Saoud, B., & Ibraheem Shayea. (2023b). Enhancing LAN Failure Predictions with Decision Trees and SVMs: Methodology and Implementation. *Electronics*, *12*(18), 3950–3950. https://doi.org/10.3390/electronics12183950

Sharma, R., Richardson, M., Martins, G., & Feamster, N. (2023). Measuring the Prevalence of WiFi Bottlenecks in Home Access Networks. *ArXiv (Cornell University)*. https://doi.org/10.48550/arxiv.2311.05499

Tan, T., Tang, F., Luo, L., Wang, X., Li, Z., & Zhao, M. (2025). *Adapting Network Information into Semantics for Generalizable and Plug-and-Play Multi-Scenario Network Diagnosis*. ArXiv.org. https://arxiv.org/abs/2501.16842

Tuli, S., Casale, G., & Jennings, N. R. (2022, August 16). *DRAGON: Decentralized Fault Tolerance in Edge Federations*. https://doi.org/10.48550/arXiv.2208.07658

Zhao, L., He, C., & Zhu, X. (2024). A Fault Diagnosis Method for 5G Cellular Networks Based on Knowledge and Data Fusion. *Sensors*, *24*(2), 401–401. https://doi.org/10.3390/s24020401